

CL420: Introduction to Biochemical Engineering

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Quiz 2 Solution Sheet

1. The 81-fold rule states that $s_{0.9}/s_{0.1} = 81^{(1/n)}$ regardless of K . Explain why K drops out, and what this tells you about the roles of n and K in the design of a biological switch. [1 mark]

Solution: Write the Hill equation in log form: $\log[\theta/(1-\theta)] = n \log s - n \log K$. Taking the ratio of the two substrate concentrations at $\theta = 0.9$ and $\theta = 0.1$ subtracts the two expressions, and the $-n \log K$ term is identical in both, so it cancels identically.

The result $s_{0.9}/s_{0.1} = 81^{(1/n)}$ contains no K at all.

Interpretation: K sets where the switch sits, n sets how sharp it is — the two are completely orthogonal design variables. To move the threshold you change affinity; to sharpen the response you change cooperativity. This is exactly why allosteric effectors that shift K (K -systems) can reposition a switch without touching its steepness.

2. A biochemist reports that in her pathway, enzyme E_3 has the lowest $V_{\max}$, and therefore concludes E_3 is "the rate-limiting step and the obvious overexpression target." Critique this conclusion using MCA language. What measurement would you ask her to make instead? [1 mark]

Solution: $V_{\max}$ is a local property of an isolated enzyme; flux control is a systemic property of the enzyme embedded in the pathway. There is no theorem connecting them. An enzyme with a low $V_{\max}$ may sit far from saturation with a large elasticity, and high elasticity generally goes with low control — the enzyme absorbs perturbations in its own substrate pool rather than transmitting them to the flux. Conversely an enzyme with a high $V_{\max}$ may hold most of the control. If $C^J_{E3} \approx 0$, overexpressing $E3$ will barely move the flux however much activity is added. The right measurement is the flux control coefficient: perturb $E3$ (titration, inducible promoter, specific inhibitor) and measure $\partial \ln J / \partial \ln E3$ — or measure the elasticities in vitro and obtain the control coefficients from the connectivity and summation theorems.

3. An enzyme can exist in three forms depending on pH: a fully protonated form (E), a singly deprotonated form (E^-), and a doubly deprotonated form (E^{2-}), but only E^- is catalytically active. Why does this setup produce a bell-shaped curve when you plot activity against pH, instead of activity just steadily rising or falling as pH changes? [1 mark]

Solution: At low pH, there's plenty of H^+ around, so the enzyme mostly stays "fully protonated" in the inactive E form. At high pH, there's too little H^+ to stop the second proton from leaving, so the enzyme drifts into the inactive E^{2-} form. The active E^- form can only dominate in the pH range between these two extremes — so activity climbs as you move away from very low pH, peaks in the middle, then falls again as you approach very high pH.

4. What is EC nomenclature? [1 mark]

Solution: EC nomenclature is a standardized system used to classify and name enzymes based on the chemical reaction they catalyze. Each enzyme is assigned an EC number with four parts/digits. Each digit can represent the enzyme class, type of reaction, specific enzyme name.

5. Why can't an uncompetitive inhibitor be overcome by adding more substrates? [1 mark]

Solution: The inhibitor binds only to the already-formed enzyme-substrate complex, not the free enzyme. Adding more substrate simply creates more ES complexes for the inhibitor to bind to, lowering both V_{max} and K_m .

6. What are the different mechanisms through which a cell can regulate the enzymatic reactions? [1 mark]

Solution: Inhibition, total enzyme concentration, substrate concentration, post-translational modifications, allosteric regulation

7. For an enzyme with two ionizable groups, given the active fraction values of K with $K_1 = 4 \times 10^{-3}$ and $K_2 = 2 \times 10^{-5}$.

- (a) Derive the optimum pH and evaluate it.
- (b) Calculate the maximum active fraction. Given $k \cdot e_{tot} = 0.30 \text{ h}^{-1}$, find V_{max} at the optimum. [2 marks]

Ans>> (a) Derive and evaluate the optimum pH

Active fraction expression:

$$f = \frac{1}{1 + \frac{[H^+]}{K_1} + \frac{K_2}{[H^+]}}$$

The denominator is minimized when:

$$\frac{[H^+]}{K_1} = \frac{K_2}{[H^+]}$$

So:

$$[H^+]^2 = K_1 K_2$$

$$[H^+] = \sqrt{K_1 K_2}$$

Substitute values:

$$[H^+] = \sqrt{(4 \times 10^{-3})(2 \times 10^{-5})}$$

$$[H^+] = \sqrt{8 \times 10^{-8}} = 2.83 \times 10^{-4}$$

$$pH = -\log_{10}(2.83 \times 10^{-4}) \approx 3.55$$

Optimum pH ≈ 3.6

(b) Maximum active fraction and V_{max}

At optimum, the two terms are equal, so:

$$D = 1 + 2 \cdot \frac{[H^+]}{K_1}$$

$$\frac{[H^+]}{K_1} = \frac{2.83 \times 10^{-4}}{4 \times 10^{-3}} = 0.0708$$

$$D = 1 + 2(0.0708) = 1.142$$

$$f_{max} = \frac{1}{1.142} = 0.875$$

Maximum active fraction ≈ 0.88

Given: $k \cdot e_{tot} = 0.30 \text{ h}^{-1}$

$$\begin{aligned}V_{max} &= f_{max} \cdot (k \cdot e_{tot}) \\V_{max} &= 0.875 \cdot 0.30 = 0.263 \text{ h}^{-1} \\V_{max} &\approx 0.26 \text{ h}^{-1}\end{aligned}$$

Final Results

- (a) Optimum pH ≈ 3.6
- (b) Maximum active fraction ≈ 0.88
- $V_{max} \approx 0.26 \text{ h}^{-1}$

8. For a microorganism, where the maintenance is given as $m_s = 0.040 \text{ g (g DW} \cdot \text{h)}^{-1}$ with $Y_{xs}^{\text{true}} = 2.0 \text{ g/g}$.

- (a) Calculate the observed yield $Y_{x/s}$ at $\mu = 0.10 \text{ h}^{-1}$ and at $\mu = 0.40 \text{ h}^{-1}$.
- (b) Using $x = D(s_f - s)/(Y_{xs}^{\text{true}} \cdot D + m_s)$, calculate x at $D = 0.10 \text{ h}^{-1}$ with $s_f = 10 \text{ g/L}$ (take $s \approx 0$). [2 mark]

Ans>> (a) Observed yield $Y_{x/s}$ at different growth rates

Using **Pirt's maintenance relation**:

$$\frac{1}{Y_{x/s}} = \frac{1}{Y_{x/true}} + \frac{m_s}{\mu}$$

- At $\mu = 0.10 \text{ h}^{-1}$:

$$\begin{aligned}\frac{1}{Y_{x/s}} &= \frac{1}{2.0} + \frac{0.040}{0.10} = 0.5 + 0.4 = 0.9 \\Y_{x/s} &\approx 1.11 \text{ g/g}\end{aligned}$$

- At $\mu = 0.40 \text{ h}^{-1}$:

$$\begin{aligned}\frac{1}{Y_{x/s}} &= \frac{1}{2.0} + \frac{0.040}{0.40} = 0.5 + 0.1 = 0.6 \\Y_{x/s} &\approx 1.67 \text{ g/g}\end{aligned}$$

Interpretation: The observed yield rises toward the true yield ($Y_{x/true} = 2.0$) as μ increases because maintenance costs become a smaller share of total costs $1/Y_{xs} = 1/Y_{xs}^{\text{true}} + m_s/\mu$ at higher growth rates.

(b) Biomass concentration at steady state

Formula given:

$$x = \frac{D(s_f - s)}{Y_{x/true} \cdot D + m_s}$$

Substitute values:

- $D = 0.10 \text{ h}^{-1}$
- $s_f = 10 \text{ g/L}$, $s \approx 0$
- $Y_{x/true} = 2.0 \text{ g/g}$
- $m_s = 0.040 \text{ g/(g} \cdot \text{h)}$

$$\begin{aligned}x &= \frac{0.10 \cdot 10}{2.0 \cdot 0.10 + 0.040} \\x &= \frac{1.0}{0.20 + 0.040} = \frac{1.0}{0.240} \approx 4.17 \text{ g/L}\end{aligned}$$

Final Results

- (a) $Y_{x/s} \approx 1.11 \text{ g/g}$ at $\mu = 0.10 \text{ h}^{-1}$

- (a) $Y_{x/s} \approx 1.67 \text{ g/g}$ at $\mu = 0.40 \text{ h}^{-1}$
- (b) Biomass concentration $x \approx 4.17 \text{ g/L}$ at $D = 0.10 \text{ h}^{-1}$